Anqi Xu

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Staff Computer Vision & Robotics Engineer with 15+ years turning perception and control algorithms into production on real hardware, from Ph.D. research in human-robot collaboration to Meta-scale XR deployment and architecting AR perception systems for frontline and enterprise users. Sets cross-team technical direction, coaches engineers into owners, and builds AI-native guardrails (context, skills, tests) that accelerate projects and teams.

WORK EXPERIENCE

Rivet Industries — 08/2026 - Present
Software Engineer, Perception Team Lead — Bellevue, WA

- Perception systems architect: Own architectures for rendering, tracking, calibration, and latency metrology across **3 orgs**; coached junior owners to reduce AR display distortion **10×**, and to automate through-the-lens rendering latency tests at **20× lower hardware cost**.

  Tech: VIO, outside-in constellation tracking, photodiode photon-to-photon rig, robotic motion-to-photon setup.
- Factory calibration: Exposed **thermal IMU clock-rate errors** the incumbent vendor system missed and mitigated a **warped fiducial target** via per-observation 3D pose modeling; selected by the C-suite to own AR device factory calibration, leading 5 engineers, 4 partners, and 11 vendor staff and keeping bring-up on schedule across 10+ risks.

  Tech: Camera/IMU/magnetometer/LED calibration, Kalibr, multi-sensor clock sync, IP licensing.
- Simulation-first agentic engineering: As the camera+IMU+display calibration tech lead, built a visual-physics simulator whose ground truth gated calibration CI and validated long-running coding agents, beating the vendor's real-world calibration results in **1 month** (**10× lower extrinsic error** vs CAD, **2× lower reprojection RMS**).

  Tech: Panda3D, Claude Code, Codex, Architecture Decision Records, harness-agnostic agent skills.
- Engineering leadership: Coached **4 direct reports** on careers and projects; mentored 2 partner-org engineers; as hiring manager for 4 roles, rewrote 2 JDs and interviewed **30+ candidates**; launched a company-wide "Best Agentic Dev Practices" series (15 per session).

Meta (Reality Labs) — 02/2021 - 07/2026
Senior Computer Vision Engineer — Redmond, WA

- Production vision systems: Helped move a product launch metric from **57% below to 42% above the market leader** by building a hardware-in-the-loop AR/MR perceptual-metrology system, adopted as the **org-wide standard**, that combined robot arms, **50-Gbps multi-camera ingestion**, visual localization, and world-locked rendering.

  Tech: C++, Python, OpenCV, OpenXR, sub-ms sensor/display synchronization, robot-arm trajectory design.
- Factory-scale metrology: Deployed calibration and validation across **6 AR/MR programs**, cutting error **up to 70%** via geometric priors and tuned fiducials and run-to-run variance **10×**; held accuracy across device variations in factory operation via GR&R acceptance limits, thermal-drift modeling, and root-cause analysis.

  Tech: Multi-view camera/IMU/display calibration, bundle adjustment, Ceres, fiducial detection, telemetry.
- Agentic engineering: Compressed a factory build from **6 weeks to 2.5 weeks** using spec-driven development, automated release packaging, and deterministic validation; built reusable agent skills adopted across **20 engineers and 3 partner teams**.

  Tech: Context/memory management, multi-agent orchestration, agent skills development, HIL debugging.
- Modular ingestion platform: Removed an extra PC hop from AR/MR data capture by replacing legacy transport with native Android modules, simplifying deployment and debugging; adopted by **4 teams across 3 orgs**.

  Tech: gRPC/protobuf, publisher/subscriber, Android, Linux, Windows.

- Performance & code health: Cut a critical-path system from **140 to 17 seconds without accuracy loss** via parallelism and pipelining with end-to-end runtime regression tests; led a dependency-safe monorepo modernization that removed **2M+ lines of dead code** and raised team test coverage from **23% to 68%**.

  Tech: C++, Python, multithreading, performance profiling, hermetic builds, Docker, CI/CD.
- Technical leadership: Owned architecture and roadmap across **2 production workstreams over 5 years**; led 4 engineers through releases aligned with feature, device, and factory partners; mentored interns, contributing to **2 full-time conversions**.

Element AI 03/2017 - 12/2020
Applied Research Scientist, Robotics Lead Montreal, Canada

- Deep reinforcement learning: Led an applied ML team building scaled infrastructure to train **PPO→SAC** policies in a high-fidelity sailing simulator, extended to RL vehicle co-design; learned strategies contributed to Emirates Team New Zealand's **2021 America's Cup win**.

  Tech: Ray/Tune/RLlib, PyTorch, Docker, Python, AWS SageMaker/S3, large-scale experiment orchestration.
- Sim-to-real robustness: Published (**ICCV 2019**) a method for generating inconspicuous adversarial images purely in simulation; verified transfer by diverting visual trackers off target in real-world monitoring and drone-photography setups.

  Tech: PyTorch, adversarial ML, domain randomization, Gazebo, ROS, OpenCV, visual servoing.
- 3D visual inspection: Demonstrated an anomaly-localization prototype at a **NeurIPS 2019** industry booth, showcasing few-shot inspection of natural objects by combining RGB-D reconstruction, object-pose regression, and learned visual embeddings.

  Tech: PyTorch, Open3D, RGB-D cameras, instance segmentation, contrastive learning.

SELECTED PROJECTS

- ueye_cam: Open-source C++/ROS driver for IDS cameras (**100+ GitHub forks**) enabling synchronized, high-throughput multi-camera capture for robotics/video analytics.
- Overcooked Bot: Built an interactive game-playing agent spanning CNN→ViT detection, appearance/Kalman tracking, homography localization, PID control, and task planning; developed a PyQt tool for model-bootstrapped data labeling. Demos: detection, tracking.
- Aqua robot platform: Refactored a real-time motor-control API from instantaneous joint commands to smooth, efficient parametric gaits; integrated IMU/GPS/camera/display drivers, embedded electronics, and ROS services; led pool and field trials.

SELECTED PUBLICATIONS

- A. Xu and G. Dudek. Maintaining Efficient Collaboration with Trust-Seeking Robots. IROS 2016. **Finalist, IROS KROS Best Paper Award on Cognitive Robotics.**
- D. Meger, J. Gamboa, A. Xu, P. Giguère, and G. Dudek. Learning Legged Swimming Gaits from Experience. ICRA 2015. **Finalist, ICRA Best Conference Paper Award.**

EDUCATION

Ph.D. in Computer Science McGill University, Canada • 2017
Thesis: Efficient Collaboration with Trust-Seeking Robots
Vanier Canadian Graduate Scholar • CIPPRS Doctoral Dissertation Award (2016)

B.Eng. in Computer Engineering McGill University, Canada • 2008
Great Distinction • Dean's Honour List

LEADERSHIP & COMMUNITY

- Program co-chair, Conference on Computer and Robot Vision (CRV), 2018-2019.
- Reviewer for ICRA, IROS, RSS, RA-L, HRI, CoRL, CRV, ICCV, and IJRR.